Type-I calibration of fixed-effect interaction tests in linear mixed models: a staged diagnosis of working-covariance mis-specification and a cluster-robust remedy

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2026-06-10

Contents

Abstract

Background. The linear mixed model is the standard tool for testing a fixed-effect interaction in longitudinal and clustered data. The model-based Wald test of that interaction is exact only when the working within-cluster covariance is correctly specified. In practice an analyst chooses a convenient parametric residual structure, a continuous-time autoregressive correlation, compound symmetry, or an unstructured form, and then trusts the model-based standard error. The direction and the magnitude of the mis-calibration that follows when the working covariance is wrong are rarely checked.

Methods. We set out a staged diagnostic protocol, a four-channel decomposition of the test statistic, a working-covariance model ladder, and a sample-size gradient, and apply it to a worked example: the test of a biomarker-by-treatment interaction in an aggregated N-of-1 carryover simulation, analysed with an `lme` model carrying a random intercept and a corCAR1 residual correlation.

Results. At the null the model-based standard error of the interaction is over-estimated by 6 to 10 percent, and the realised type-I error is approximately 0.03 against a nominal 0.05. The degrees-of-freedom rule, the positive-definiteness correction, and point-estimate bias are excluded as causes. The model ladder localises the mis-calibration to the corCAR1 residual layer: a random intercept alone is well calibrated. The sample-size gradient shows the bias is asymptotic, not a moderate-sample artefact: it does not attenuate from $N = 35$ to $N = 280$. A CR2 bias-reduced cluster-robust standard error restores the standard-error ratio to approximately one and the type-I error to approximately 0.05. At an alternative cell with a non-null effect the recalibration raises power by a near-uniform increment and leaves the ranking of analysis specifications unchanged.

Conclusions. A mis-specified parametric working covariance can bias the model-based standard error of a mixed-model interaction test asymptotically, in either direction. Cluster-robust standard errors are a general remedy, and the staged diagnostic is a reusable instrument for detecting the problem.

1 Introduction

[1]

The Wald test of a mixed-model interaction is exact only when the working covariance is correctly specified.

- The estimand is a fixed-effect interaction coefficient.
- Inference proceeds by dividing the estimate by a model-based standard error and referring to a t distribution.
- Exactness requires the assumed random-effects structure and residual correlation to match the truth.

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The linear mixed model is, by a wide margin, the most common vehicle for testing whether an effect is moderated by a covariate in longitudinal or clustered data[1,2]*. The quantity of interest is a fixed-effect interaction coefficient, and the inference almost always proceeds through a Wald test: the coefficient estimate is divided by a model-based standard error and referred to a t distribution. That procedure is exact only under a condition that is easy to state and, unfortunately, easy to forget. The working covariance, that is, the random-effects structure together with the parametric residual correlation the analyst has assumed, must be correct.*

[2]

Covariance mis-specification renders the model-based standard error inconsistent, motivating the cluster-robust sandwich estimator.

- Eicker, Huber, and White established that a mis-specified working covariance yields an inconsistent model-based standard error.
- Liang and Zeger introduced GEE, decoupling point estimation from variance estimation via an empirical sandwich.
- A substantial literature has since refined the sandwich estimator for the small-sample regime.

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It has been understood since the work of Eicker, Huber, and White that this condition is more demanding than it appears[3–5]*. When the working covariance is mis-specified, the model-based standard error is no longer a consistent estimator of the true sampling variability of the fixed-effect estimate. Liang and Zeger*[6] *carried this insight into longitudinal data analysis: their generalized estimating equations decouple the point estimate, computed under a convenient working correlation, from the variance estimate, computed by an empirical sandwich that remains consistent whatever the working correlation happens to be. The sandwich, or cluster-robust, variance estimator has since become a standard instrument, and a substantial literature has refined it for the small-sample case*[7–10]*.*

[3]

The direction and persistence of mis-calibration from a plausible but incorrect working covariance are rarely audited and cannot be answered in the abstract.

- Analysts routinely adopt parametric residual correlations on substantive grounds and trust the resulting standard errors.
- The magnitude, direction, and sample-size behaviour of the resulting mis-calibration depend on the specific interaction and covariance mismatch.
- Answering these questions requires examining a concrete analysis.

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Yet in routine mixed-model practice the working covariance is still trusted. An analyst fitting longitudinal data with `nlme` *or a comparable package will choose a residual correlation, very often a continuous-time first-order autoregressive form, because it is pharmacokinetically or biologically plausible*[11]*, and will then report the model-based standard error of the interaction without asking what happens if the chosen form is not exactly the form the data follow. This is not negligence so much as a reasonable working assumption that is seldom audited. But how large is the error when the assumption fails? Is the resulting test conservative or anti-conservative? Does the mis-calibration diminish as the sample grows, so that it can perhaps be dismissed as a small-sample nuisance, or does it persist? These questions do not have a single answer, because the answer depends on the interaction being tested and on the way the working covariance departs from the truth. They have to be asked of a particular analysis, and we shall ask them of one.*

[4]

This paper contributes a reusable four-channel diagnostic protocol and a worked finding of asymptotic, conservative mis-calibration in a longitudinal interaction test.

- The protocol decomposes the Wald statistic into bias, scale, degrees of freedom, and reference-shape channels, supplemented by a model ladder and a sample-size gradient.
- The worked example reveals a conservative test whose standard error is over-estimated by roughly ten percent.
- The conservatism is asymptotic and is removed by a cluster-robust standard error.

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In this paper we endeavour to do exactly that for one analysis, and in doing so we offer two contributions that we believe are more general than the single case. The first is a diagnostic protocol. We decompose the Wald statistic into four channels through which mis-calibration can enter, and we add two further instruments: a working-covariance model ladder that localises the cause, and a sample-size gradient that separates a finite-sample artefact from an asymptotic one. The protocol is not specific to the example and can be applied to any mixed-model interaction test whose calibration is in doubt. The second contribution is the finding the protocol produced. In our worked example the model-based interaction test is conservative, not anti-conservative as the sandwich literature's emphasis on under-estimated standard errors might lead one to expect; the conservatism is material, on the order of ten percent in standard-error terms; and it is asymptotic, persisting undiminished across an eightfold range of sample size. We trace it to a single component of the working covariance, and we show that a cluster-robust standard error removes it.

[5]

The worked example is drawn from a carryover simulation study and serves solely to make the general phenomenon concrete.

- The example is the biomarker-treatment interaction test in the pmsimstats-ng carryover-sensitivity study.
- The finding is a property of the mixed-model Wald test, not of the particular simulation.
- Sections 2 through 5 address the protocol, the example, the staged results, and their generalisation.

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The worked example is drawn from the pmsimstats-ng project, a simulation study of power to detect biomarker-treatment interactions in aggregated N-of-1 trials under carryover[12]*. The example is incidental to the argument. The phenomenon, a parametric working covariance that biases a fixed-effect interaction test asymptotically, and the remedy are not properties of that simulation; they are properties of the mixed-model Wald test, and the example* *serves only to make them concrete and quantified. We begin in Section 2 with the diagnostic protocol. Section 3 describes the worked example. Section 4 reports the staged diagnosis and the cluster-robust remedy. Section 5 discusses the generality of the result and its limits.*

2 A staged diagnostic protocol

[6]

Mis-calibration of a Wald test must enter through exactly one of four separable channels, each diagnosable by simulation.

- The test statistic is $T = \hat{\theta}/\widehat{\mathrm{SE}}$, referred to a t_ν critical value.
- Correct calibration requires the realised rejection rate to equal α under H_0.
- The four channels – bias, scale, degrees of freedom, and reference shape – are mutually exhaustive and separable.

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Consider a fixed-effect interaction coefficient θ estimated by $\hat{\theta}$, with a model-based standard error $\widehat{\mathrm{SE}}$ and an assigned degrees of freedom ν. The Wald test rejects the null $H_0 : \theta = 0$ at level α when the statistic $T = \hat{\theta}/\widehat{\mathrm{SE}}$ exceeds the critical value $t_{\nu,1-\alpha/2}$ in absolute value. The test is correctly calibrated when the realised rejection rate under H_0 equals α. When it does not, the discrepancy must enter through one of exactly four channels, and the channels are separable by simulation. The general anatomy of a Wald statistic and of the reference distribution against which it is judged is itself classical[13]. We begin with the decomposition itself, and then describe the two further instruments we shall add to it.

2.1 The four-channel decomposition

We begin by writing the interaction estimate in a form that exposes the four channels. We use the word 'channel' for the four mutually exclusive routes by which a discrepancy can reach the realised error rate; the term is adopted because the more familiar nouns are already committed in this setting, 'component' to variance components, 'factor' to the latent factors of the data-generating process, and 'term' to the terms of the model. Let

$$\hat{\theta} \;=\; \mu + \sigma\, Z^{\star}, \qquad \mu = \mathrm{E}(\hat{\theta}), \quad \sigma = \mathrm{SD}(\hat{\theta}), \tag{1}$$

so that $Z^{\star} = (\hat{\theta} - \mu)/\sigma$ is the standardised estimate, of mean zero and unit variance by construction; when $\hat{\theta}$ is approximately normal, $Z^{\star}$ is approximately standard normal. Write $\overline{\widehat{\mathrm{SE}}}$ for the mean of the model-based standard error and $\kappa = \overline{\widehat{\mathrm{SE}}}/\sigma$ for its ratio to the true sampling standard deviation σ. Then the Wald statistic, to the order at which the variability of $\widehat{\mathrm{SE}}$ about its mean may be neglected, is

$$T \;=\; \frac{\hat{\theta}}{\overline{\widehat{\mathrm{SE}}}} \;\approx\; \frac{\mu + \sigma Z^\star}{\kappa\,\sigma} \;=\; \frac{1}{\kappa}\,(Z^\star + \delta), \qquad \delta = \frac{\mu}{\sigma}, \tag{2}$$

where δ is the bias expressed in standard-deviation units. In this form the statistic is a shifted and rescaled standard normal, the structure that underlies the non-central t distribution[14]: δ enters as a non-centrality and κ as a scale distortion. The realised type-I error follows at once,

$$\alpha_{\text{real}} \;=\; \Pr\!\big(|T| > t_{\nu,\,1-\alpha/2}\big) \;\approx\; \Pr\!\big(|Z^\star + \delta| > \kappa\, t_{\nu,\,1-\alpha/2}\big), \tag{3}$$

and exactly four quantities appear in it. They are the four channels: the standardised bias δ, which decentres the statistic; the scale κ, which multiplies the rejection threshold; the degrees of freedom ν, which fix the critical value $t_{\nu,1-\alpha/2}$; and the distribution of $Z^\star$, the shape of the reference. When all four take their nominal values, $\delta = 0$, $\kappa = 1$, $\nu = \infty$, and $Z^\star \sim \mathrm{N}(0,1)$, the expression collapses to $\Pr(|\mathrm{N}(0,1)| > z_{1-\alpha/2}) = \alpha$ and the test is exact; a departure in any one channel mis-calibrates it. A second identity is worth recording, since the diagnosis turns on it: because $T \approx (Z^\star + \delta)/\kappa$, the dispersion of the statistic satisfies

$$\mathrm{SD}(T) \;\approx\; \mathrm{SD}(Z^\star)\,/\,\kappa, \tag{4}$$

so a statistic standard deviation below one is read at once as a scale inflation ($\kappa > 1$), a sub-normal reference shape, or both. One qualification attends the decomposition: equation (2) treats κ as a fixed mean ratio and neglects the sampling variability of $\widehat{\mathrm{SE}}$ itself. That variability is not lost but carried by the degrees-of-freedom and reference-shape channels, with which the scale channel is therefore only approximately separable. The approximation is excellent when the model assigns many degrees of freedom to the standard error, as in the worked example below, and we return to its limits where the cluster-robust standard error, which carries far fewer, is taken up.

We measure the four channels by simulation as follows.

1. **Point-estimate bias** (δ). If $\mathrm{E}(\hat{\theta}) \neq 0$ under H_0 the statistic is decentred, acquiring the non-centrality of a noncentral t[14]. We measure μ, and hence $\delta = \mu/\sigma$, by the mean of $\hat{\theta}$ across null replicates.
2. **Standard-error calibration** (κ). If the model-based $\widehat{\mathrm{SE}}$ does not estimate the true sampling standard deviation of $\hat{\theta}$, the statistic is rescaled. We measure κ as the mean model-reported standard error over the empirical standard deviation of the estimate across replicates. That unequal error variances or correlation among the errors distort the realised level of a test through this channel is a classical result of Box[15,16]. A ratio above one indicates an over-stated standard error and a conservative test; a ratio below one the reverse.

3. **Degrees of freedom** (ν). If ν is too small the t reference has tails too heavy and the test is conservative; if too large, the reverse. The approximate degrees of freedom such a reference carries trace to Satterthwaite[17] and Welch[18]. We measure this by recomputing the p-values against a normal reference and comparing the realised type-I error.
4. **Reference-distribution shape** (the law of $Z^\star$). Even with $\delta = 0$, $\kappa = 1$, and ν correct, $Z^\star$ may not be standard normal. The consequences of such non-normality for the true level of a test were studied classically by Pearson[19], Box[20], and Scheffé[21, ch. 10]. We measure this by the standard deviation of T under H_0, which should be approximately one, and by the empirical distribution function of the p-values, which should be $\text{Uniform}(0, 1)$.

[7]

The decomposition is novel not for introducing new measures but for showing that exactly these four quantities compose into the realised type-I error, making the diagnosis exhaustive and localising.

- Each channel – bias, scale, degrees of freedom, shape – has a classical antecedent: the noncentral t of Johnson and Welch, the level distortions from unequal variance and correlation of Box, the approximate degrees of freedom of Satterthwaite and Welch, and the non-normality studies of Pearson, Box, and Scheffé.
- Together they are exhaustive: a discrepancy absent from all four cannot exist.
- Because the channels enter the expression in distinct, non-interacting roles, the decomposition localises the fault rather than merely detecting it.

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The individual channels are not new, and each carries a classical provenance. The decentering of the statistic by a non-null mean is the non-centrality of the noncentral t distribution, whose properties were charted by Johnson and Welch[14]; as performance measures, the bias of the estimate and the ratio of the model-based to the empirical standard error are among the standard summaries of a simulation study codified by Morris, White, and Crowther[22], who recommend that the model-based and empirical standard errors be reported and compared as a matter of course. The scale channel has the longest pedigree of the four: that unequal error variances and correlation among the errors push the realised significance level of a test away from its nominal value, precisely the mechanism by which a mis-specified working covariance *acts here, was established by Box in his two-part study of departures from the assumptions of the analysis of variance[15,16], the asymptotic counterpart being the inconsistency of the model-based standard error under the conditions of Eicker, Huber, and White[3–5]. The degrees-of-freedom channel rests on the approximate-distribution arguments of Satterthwaite[17] and Welch[18], refined for the mixed model by Kenward and Roger[23]. The reference-shape channel, the departure of the law of $Z^\star$ from normality, is the subject of the classical robustness literature of Pearson[19], Box[20], and Scheffé[21, ch. 10], its formal small-sample anatomy given by the Edgeworth expansion of the studentised statistic[24], with the uniformity of the null p-value distribution following from the probability integral transform; the anatomy of the Wald statistic that organises the four is itself standard[13]. What the decomposition contributes is not a new measure, nor the observation that any single assumption matters, but the demonstration that these four quantities, and exactly these four, compose into the single expression for α_{real} derived above. Two consequences follow. First, the channels are exhaustive: a Wald test is a centred statistic, a scale, a reference shape, and a degrees-of-freedom choice, and the realised error rate is a function of those and of nothing else, so a discrepancy absent from all four cannot exist. Second, because the channels enter the expression in distinct and non-interacting roles, measuring all four does not merely detect a mis-calibration but localises it, identifying which term carries the fault. Since the channels have different causes and different remedies, a diagnosis is in this sense a decomposition and not a checklist.*

2.2 The working-covariance model ladder

The four-channel decomposition tells us how a test is mis-calibrated. It does not tell us why. When the decomposition points to the standard-error channel, it is worth being precise about what the scale κ measures. Let the interaction estimate be written $\widehat{\theta} = \ell'\widehat{\beta}$, with ℓ the contrast vector that selects the interaction coefficient from the fixed-effects vector. A fixed-effect estimator that solves a generalized least squares system under a working covariance V_0, when the data are in truth generated under a covariance V, has a model-based variance

$$v_0 \;=\; \ell'\big(X'V_0^{-1}X\big)^{-1}\ell, \tag{5}$$

while its true sampling variance is the sandwich form

$$v \;=\; \ell'\big(X'V_0^{-1}X\big)^{-1}\big(X'V_0^{-1}VV_0^{-1}X\big)\big(X'V_0^{-1}X\big)^{-1}\ell, \tag{6}$$

where X is the fixed-effects design and, the clusters being independent, every cross-product is a sum over clusters, $X'V_0^{-1}X = \sum_i X_i'V_{0i}^{-1}X_i$ and likewise $X'V_0^{-1}VV_0^{-1}X = \sum_i X_i'V_{0i}^{-1}V_iV_{0i}^{-1}X_i$[5,6,25]. For the mixed model fitted by maximum or restricted maximum likelihood, the inconsistency of the model-based variance v_0 for the true variance v under a wrong V_0 is the misspecified-likelihood result of White[26], of which the sandwich form is the consistent counterpart. The scale of Section 2.1 is the ratio of these two quadratic forms, $\kappa^2 = v_0/v$. They coincide when $V_0 = V$, and for a generic contrast ℓ they do not otherwise; the scale channel is therefore not a finite-sample nuisance but a direct consequence of the working covariance: κ departs from one whenever $V_0 \neq V$, and no quantity of data removes the discrepancy.

[8]

A model ladder localises the covariance component responsible for standard-error mis-calibration by tracking κ across a graded sequence of working covariances fitted to the same data.

- Each rung substitutes a different working covariance V_0 for the identical simulated data.
- The rung at which κ departs from one identifies the responsible component.
- Fitting the same data at every rung eliminates Monte Carlo noise between models.

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This is also what makes a model ladder informative. Each rung is a different working covariance V_0 fitted to the same data, so tracking κ along a graded sequence of working covariances, from the simplest to the production model, isolates the rung, that is, the covariance component, at which κ departs from one. Because every rung is fitted to the identical simulated data, the comparison is free of Monte Carlo noise between models.

2.3 The sample-size gradient

[9]

The sample-size gradient distinguishes a finite-sample artefact, which attenuates with N, from an asymptotic bias, which persists, and the distinction determines the appropriate remedy.

- A finite-sample artefact diminishes as variance parameters are estimated more precisely; Kenward-Roger corrections address this case.
- An asymptotic bias is a property of the working covariance mismatch and does not attenuate with sample size.
- A flat calibration profile across a range of N values identifies the asymptotic case.

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A standard-error mis-calibration may be one of two qualitatively different things. It may be a finite-sample artefact, in which case it attenuates as the sample grows and the working-covariance parameters are estimated more sharply; the small-sample corrections of Kenward and Roger[23,27] address mis-calibration of this kind. Or it may be asymptotic, a consequence of a working covariance that does not match the data-generating covariance no matter how much data is collected, in which case it persists. The two are distinguished by running the decomposition across a range of cluster counts. Here a point of construction guides the reading. The scale κ of Section 2.2 is a functional of the per-cluster quantities (X_i, V_i, V_{0i}) alone, so it does not depend on the number of clusters and is, to leading order, constant along the gradient by construction; what a varying cluster count changes is the precision with which the variance parameters of V_0 are estimated, the very quantity whose finite-sample uncertainty the Kenward-Roger correction addresses, following Kackar and Harville[28]. A gradient that is flat in the cluster count therefore does two things at once: it confirms that no spurious trend enters through parameter estimation, and it shows that the bias is not of the kind a finite-sample correction would remove. Attenuation toward nominal calibration marks the finite-sample case; a flat profile marks the asymptotic case. The distinction matters, because a finite-sample artefact at a realistic sample size may be tolerable, whereas an asymptotic bias is a property of the method and will not be outgrown. The gradient varies the number of clusters and not the number of timepoints within a cluster; the latter is the axis along which the per-cluster covariance mismatch could itself change, and we return to it among the limitations.

3 Worked example: an interaction test under carry-over

[10]

The worked example is a biomarker-by-treatment interaction test from an aggregated N-of-1 carryover simulation study, reported in ADEMP structure.

- Participants are repeatedly crossed between active treatment and control in an N-of-1 design.
- The scientific question is whether a baseline biomarker moderates the treatment effect.
- Carryover complicates the analysis and the study compares strategies for accommodating it.

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The example we shall work through is the test of a biomarker-by-treatment interaction in the carryover-sensitivity simulation study of the pmsimstats-ng project. Aggregated N-of-1 trials repeatedly cross each participant between an active treatment and a control condition, and the scientific question is whether a baseline biomarker moderates the treatment effect. Carryover, the persistence of drug effect into a washout period, complicates the analysis, and the study compares analysis-model strategies for accommodating it[22]*. The reporting follows the ADEMP structure of Morris, White, and Crowther*[22].

3.1 The analysis model

[11]

The analysis model is an `nlme::lme` with a random intercept and corCAR1 residual correlation, and the calibration of its interaction test is the question under examination.

- The model includes a participant random intercept and a continuous-time AR(1) residual correlation.
- The corCAR1 form was adopted for clinical realism and numerical tractability.
- Whether it is well calibrated for the fixed-effect interaction test is the central question.

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For a symptom score $S_{x,it}$ on participant i at timepoint t, the analysis model is the linear mixed model

$$S_{x,it} \;=\; \beta_0 + \beta_1\,\mathrm{bm}_i + \beta_2\,t + \beta_3\,X_{it} + \theta\,(\mathrm{bm}_i \times X_{it}) + b_i + \varepsilon_{it}, \qquad (7)$$

fitted by `nlme::lme` with a participant random intercept $b_i \sim \mathrm{N}(0, \sigma_b^2)$ and a continuous-time first-order autoregressive residual correlation, corCAR1(form $=\sim t \mid$ ptID). *Here bm_i is the standardised biomarker, X_{it} a drug-exposure predictor, and θ the interaction coefficient of interest. The continuous-time autoregressive residual was adopted in the project for its clinical realism, nearby measurements being more strongly correlated than distant ones, and for the numerical range it affords*[11]*. Whether that choice* *is also well calibrated for the fixed-effect interaction test is precisely the question we shall address.*

3.2 The null configuration

[12]

The null configuration collapses the two DGP architectures to a single process, so the calibration diagnosis need not stratify by architecture.

- Calibration is a null-hypothesis property, evaluated at $c_{bm} = 0$ and $\theta = 0$.
- The primary cell is the Hybrid design at $N = 70$ with a one-week carryover half-life.
- At the null, both DGP architectures produce bit-identical data.

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Calibration is a null-hypothesis property, so the diagnosis is run where the biomarker exerts no moderation, $c_{bm} = 0$, and the true interaction is therefore $\theta = 0$. The primary cell is the reference configuration of the parent study: the Hybrid N-of-1 design, $N = 70$ participants, an exponential carryover form with a one-week half-life, and an autoregressive parameter of 0.7. A preliminary check established that at $c_{bm} = 0$ the study's two data-generating architectures collapse to a bit-identical process, so the diagnosis need not stratify by architecture.

[13]

The working covariance is structurally incapable of matching the data-generating covariance, making this a prototypical instance of the mis-specification problem.

- The simulated symptom score is a mixture of three autoregressive factors with timepoint-dependent variances plus a participant baseline.
- A single random intercept and single-parameter corCAR1 residual cannot reproduce this structure.
- The example is therefore one of plausible, conventional, and inexact working covariance specification.

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The feature of the example that merits attention for the argument is the relationship between the working covariance and the truth. The simulated symptom score is a sum of three latent factors, each following its own autoregressive process, with timepoint-dependent variances, plus a participant-level baseline term. The working covariance of the analysis model, one random intercept and one single-parameter corCAR1 residual, cannot reproduce that structure exactly. The example is therefore an instance of the general situation the protocol is designed for: a parametric working covariance that is plausible, conventional, and not exactly correct.

4 Results

[14]

Simulation runs are sized to keep Monte Carlo standard error below 0.005 for each diagnostic component, with all replicate-level outputs retained.

- The decomposition and model ladder use 5000 null replicates; the sample-size gradient uses 1500 per point.
- The cluster-robust verification uses 3000 replicates.
- At a type-I error near 0.03, the Monte Carlo standard error is approximately 0.0024 at 5000 replicates.

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The diagnosis was carried out at 5000 null replicates for the four-channel decomposition and the model ladder, and at 1500 replicates per point for the sample-size gradient. The cluster-robust remedy was verified at 3000 replicates. A rejection rate estimated from n_{sim} replicates carries a Monte Carlo standard error of $\sqrt{p(1-p)/n_{\mathrm{sim}}}$[22]; at a type-I error near 0.03 this is approximately 0.0024 at 5000 replicates, 0.0031 at 3000, and 0.0044 at 1500, and the comparisons to the nominal level reported below are read against those tolerances. All replicate-level outputs are retained for inspection.

4.1 The conservatism is standard-error over-estimation

[15]

Table 1: Four-channel decomposition at the primary null cell (Hybrid design, $N = 70$, $c_{bm} = 0$, 5000 replicates), for the three analysis specifications A1, A2, A3. The interaction estimate is effectively unbiased; the SE ratio exceeds one; the statistic is correspondingly compressed; the t and z references coincide because the degrees of freedom are large.

Spec	Mean estimate	SE ratio κ	SD(T)	Type I (t)	Type I (z)	Mean DF
A1	+0.032	1.069	0.932	0.033	0.034	487
A2	+0.039	1.097	0.909	0.029	0.029	487
A3	+0.033	1.061	0.939	0.035	0.036	486

The decomposition table excludes point-estimate bias and the degrees-of-freedom channel, leaving the standard-error channel as the site of the conservatism.

- All three specifications show a negligible null-mean displacement of about 0.035 standard deviations.
- The model assigns approximately 487 degrees of freedom; switching to a normal reference moves the type-I error by only 0.001.
- The conservatism must therefore reside in the standard-error channel.

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Table 1 reports the decomposition for the three analysis specifications carried by the parent study, which differ only in the drug-exposure predictor X_{it} and are treated here as three instances of the same interaction test. The reading is consistent across all three. The interaction estimate is effectively unbiased: the null mean falls between +0.03 and +0.04 across the three specifications, on a coefficient whose sampling standard deviation is near unity, a displacement of about 0.035 of a standard deviation. A displacement of that size is statistically detectable at 5000 replicates but is negligible in magnitude, and in any case it points toward rejection and so cannot produce a conservative test. The degrees-of-freedom channel is excluded directly: the model assigns approximately 487 degrees of freedom to the interaction term, and recomputing the p-values against a normal reference moves the realised type-I error by about 0.001, as the near-identical t- and z-reference columns show.

The discrepancy is in the standard-error channel. The model-based standard error exceeds the empirical sampling standard deviation of the estimate by 6 to 10 percent, and the standard deviation of the test statistic agrees with the reciprocal of that ratio to within half a percent, as the identity $\mathrm{SD}(T) \approx \mathrm{SD}(Z^\star)/\kappa$ of equation (4) with $\mathrm{SD}(Z^\star) = 1$ leads us to expect. With the other three channels inert, $\delta \approx 0$, ν large, and $Z^\star$ standard normal, the realised type-I error of equation (3) reduces to the closed form

$$\alpha_{\mathrm{real}} \;\approx\; 2\,\Phi\!\left(-\kappa\, t_{\nu,\,1-\alpha/2}\right), \tag{8}$$

the probability that a standard normal exceeds the threshold $\kappa\, t_{\nu,1-\alpha/2}$ rather than $z_{1-\alpha/2}$.

[16]

A first-order approximation gives the realised type-I error an elasticity of about -4.6 in the standard-error ratio at the nominal point, so a 6–10 percent over-statement is materially consequential.

- With the other three channels inert the realised error reduces to $2\Phi(-\kappa\, t_{\nu,1-\alpha/2})$.
- The derivative near $\kappa = 1$ is approximately -0.23, an elasticity near -4.6, so the type-I error falls to the low 0.03 range.
- The predicted range of 0.031 to 0.037 matches the observed 0.029 to 0.035 to within Monte Carlo error.

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Differentiating this closed form exposes the leverage of the scale channel. Near $\kappa = 1$, with ν large enough that $t_{\nu,1-\alpha/2} \approx z_{1-\alpha/2} = 1.96$,

$$\frac{\mathrm{d}\,\alpha_{\mathrm{real}}}{\mathrm{d}\,\kappa} \;=\; -2\, t_{\nu,1-\alpha/2}\, \phi\!\left(\kappa\, t_{\nu,1-\alpha/2}\right) \;\approx\; -0.23, \tag{9}$$

so to first order $\alpha_{\mathrm{real}} \approx 0.05 - 0.23\,(\kappa - 1)$. The leverage is best read as an elasticity: at $\kappa = 1$, $\mathrm{d}\log\alpha_{\mathrm{real}}/\mathrm{d}\log\kappa \approx -4.6$, so each one percent of over-statement in the standard error removes about four to five percent of the nominal type-I error. A six-to-ten percent over-statement therefore depresses the type-I error into the low 0.03 range, and it is in this sense that a numerically modest mis-calibration of the standard error constitutes a material mis-calibration of the test. For the standard-error ratios of Table 1 this predicts a type-I error of 0.031 to 0.037 across the three specifications, matching the observed 0.029 to 0.035 (at $\alpha = 0.05$) to within Monte Carlo error. The scale channel alone, with no contribution from the other three, reproduces the conservatism.

[17]

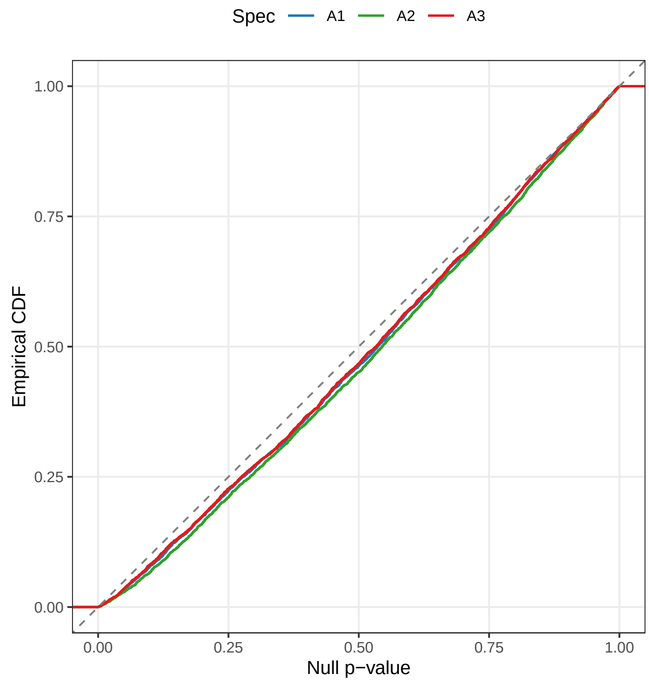

Figure 1: Empirical distribution function of the null p-values at the primary cell, against the Uniform(0,1) diagonal. All three specifications bow uniformly below the diagonal, the signature of a global scale compression rather than a tail anomaly.

The uniform bowing of the p-value CDF below the diagonal confirms global scale compression rather than a tail anomaly, and the positive-definiteness correction is excluded as a contributor.

- A tail-only failure would depart from the diagonal only near zero; the uniform bowing seen here is distinctive.
- Global scale compression is the signature of an inflated standard error.
- The positive-definiteness correction was not triggered in any of the cell's replicates.

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Figure 1 confirms the nature of the mis-calibration. The empirical distribution function of the null p-values lies below the diagonal across the whole unit interval and for all three specifications. A test that failed only in its tails would depart from the diagonal only near zero; the uniform bowing seen here is the signature of a global scale compression, which is precisely what an

Table 2: Working-covariance model ladder at the primary null cell (5000 replicates), for specifications A1 and A2. The same datasets are refitted under each structure. M1, a random intercept alone, is well calibrated; M3, the production model, adds a corCAR1 residual and is conservative.

Model	Structure	κ (A1)	κ (A2)	Type I (A1)	Type I (A2)
M0	OLS, no within-cluster correlation	1.323	1.289	0.010	0.012
M1	Random intercept only	0.989	0.972	0.049	0.055
M2	corCAR1 residual only	1.023	1.070	0.043	0.036
M3	Random intercept + corCAR1 (production)	1.068	1.097	0.033	0.029

inflated standard error produces. The positive-definiteness correction available to the data generator was not triggered in any of the cell's paths and is excluded as a contributor.

4.2 The corCAR1 residual layer is the cause

[18]

The model ladder identifies the source of the over-estimation by fitting four working covariances to the identical null datasets.

- M0 is OLS with no within-cluster term; M1 adds a random intercept; M2 adds only corCAR1; M3 combines both.
- Because every rung uses the same data, the comparison is free of Monte Carlo noise.
- The rung at which κ departs from one identifies the responsible component.

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The decomposition tells us that the standard error is over-estimated. The model ladder, Table 2, tells us which part of the model does the over-estimating. Four working covariances were fitted to the identical null datasets: ordinary least squares with no within-cluster term (M0), a random intercept alone (M1), a corCAR1 residual alone (M2), and the production combination of a random intercept with a corCAR1 residual (M3).

[19]

The corCAR1 residual layer is the proximate cause: adding it to a random intercept that is itself well calibrated removes

the calibration.

- M1 yields SE ratios of 0.97–0.99 and type-I errors of 0.049–0.055.
- M3 raises the ratio to 1.07–1.10 and depresses the type-I error to 0.029–0.033.
- The claim is not that corCAR1 is wrong absolutely but that adding it to a model that already carries a random intercept degrades calibration.

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The result is unambiguous. M1, the random intercept with no residual correlation, is well calibrated: its standard-error ratio sits between 0.97 and 0.99 and its type-I error between 0.049 and 0.055. M3, which adds the corCAR1 residual on top of that random intercept, is conservative: the ratio rises to between 1.07 and 1.10 and the type-I error falls to between 0.029 and 0.033. The corCAR1 residual-correlation layer is therefore the proximate cause of the conservatism. It is worth being precise about what this does and does not say. It does not say that the corCAR1 residual is wrong in any absolute sense; the simulated factors are genuinely autocorrelated, so a residual correlation term is capturing a real feature of the data. What it says is that adding that term to a model which already carries a random intercept degrades the calibration of the fixed-effect interaction test.

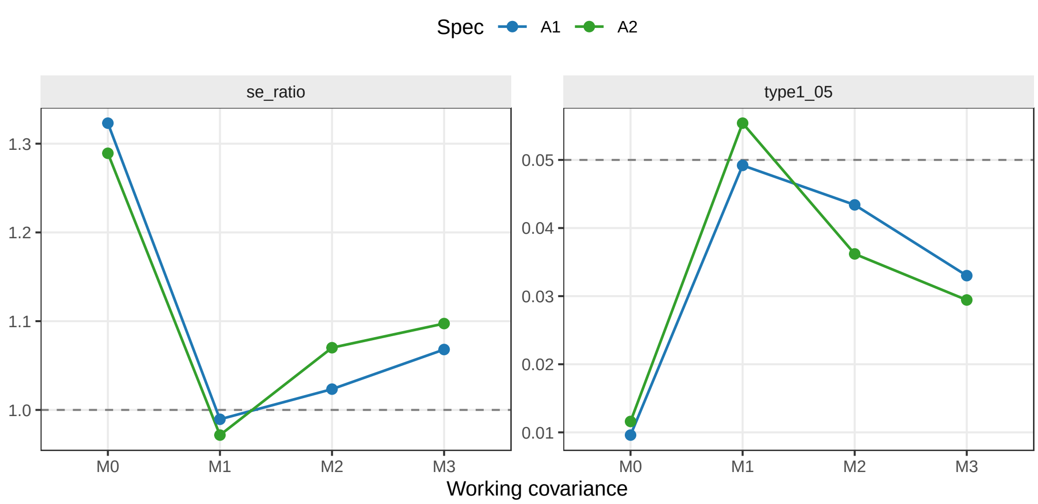

Figure 2: Interaction standard-error calibration across the model ladder. Left, the ratio of model-reported to empirical standard error; right, the realised type-I error. The dashed lines mark perfect calibration. The random intercept alone (M1) sits on both reference lines; the production model (M3) does not.

[20]

OLS is strongly conservative for within-cluster contrasts – reversing the usual Moulton intuition – and the ladder shows that the sign of the OLS error is a property of the estimand, not of ignoring correlation per se.

- OLS charges the full residual variance to a within-participant contrast when only the within-participant component is relevant, yielding $\kappa \approx 1.3$.
- The Moulton pitfall applies to between-cluster effects; for within-cluster contrasts the sign is reversed.
- M2 (corCAR1 alone) is intermediate, confirming that mis-calibration magnitude and sign depend on the working covariance choice.

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Two further features of the ladder, visible in Figure 2, are worth recording. The first is that M0, ordinary least squares, is not anti-conservative, as the familiar warning that ignoring positive within-cluster correlation understates standard errors would suggest. It is strongly conservative, with a standard-error ratio near 1.3. The reason is instructive: the interaction here is a within-participant contrast, and ordinary least squares charges it the full residual variance, between-participant and within-participant combined, when the error relevant to a within-participant contrast is only the within-participant part. In the idealised case of a pure within-participant contrast with exchangeable errors the inflation is the variance ratio $\kappa^2_{\mathrm{OLS}} \approx (\sigma_b^2 + \sigma_\varepsilon^2)/\sigma_\varepsilon^2$*, the total residual variance over its within-participant component, exceeding one by the random-intercept variance* σ_b^2*. The familiar warning, the Moulton pitfall, concerns between-cluster effects*[29]*; for a within-cluster effect the sign of the OLS error is reversed. The distinction between within- and between-cluster covariate effects that governs this sign is the one drawn by Neuhaus and Kalbfleisch*[30]*, and the inflation of the ordinary-least-squares variance under intra-cluster correlation is the two-stage-sampling result of Scott and Holt*[31]*. The second feature is that the corCAR1 residual without a random intercept (M2) is intermediate. Taken together, the ladder shows that calibration is recovered by the random intercept alone and removed by the corCAR1 addition, and that the magnitude and even the sign of the mis-calibration is a property of the working covariance, not a universal constant.*

4.3 The conservatism is asymptotic, not a small-sample artefact

[21]

The conservatism is asymptotic: the SE ratio and type-I error

Table 3: Sample-size gradient under the production model M3 (1500 replicates per point). The standard-error ratio κ and the type-I error (T1) are shown for each specification. Neither trends toward its nominal value as N increases.

N	κ A1	κ A2	κ A3	T1 A1	T1 A2	T1 A3
35	1.072	1.103	1.064	0.038	0.028	0.039
70	1.056	1.083	1.050	0.043	0.031	0.045
140	1.086	1.098	1.079	0.041	0.033	0.042
280	1.052	1.070	1.046	0.037	0.028	0.035

show no trend toward their nominal values across an eightfold range of sample size.

- From $N = 35$ to $N = 280$ the SE ratio holds between 1.05 and 1.10 with no downward trend.
- At $N = 280$ the type-I error is still 0.028, well below the nominal 0.05.
- The conservatism is a property of the M3 working covariance and will not be outgrown.

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If the conservatism were a finite-sample artefact, it would diminish as the sample grew and the working-covariance parameters were estimated more precisely. Table 3 and Figure 3 show that it does not. Across an eightfold range of sample size, from $N = 35$ to $N = 280$ participants, the standard-error ratio holds between 1.05 and 1.10 with no trend toward one, and the type-I error holds between 0.028 and 0.045 with no trend toward 0.05. At the largest sample size the production specification still rejects the true null at 0.028. The conservatism is asymptotic. It appears to be a property of the M3 working covariance and will not be outgrown.

[22]

The asymptotic bias arises because the model-based SE converges to a biased target: more data sharpens estimation of the wrong covariance, not the right one.

- The data-generating covariance is a mixture of autoregressive factors with timepoint-dependent variances that M3 cannot reproduce.
- Larger samples estimate the M3 parameters more precisely but do not correct the structural mismatch.

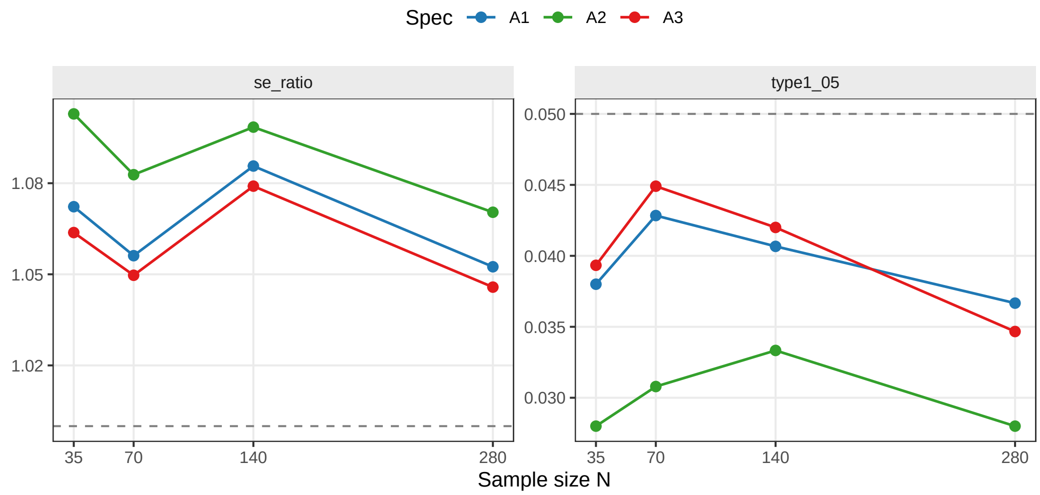

Figure 3: Type-I error and standard-error ratio against sample size under the production model M3. Neither panel trends toward its dashed reference line as the number of participants increases from 35 to 280.

- The bias is a property of the model, not of the estimation of the model's parameters.

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The explanation that is consistent with all three results is a working-covariance mismatch rather than a small-sample effect. The simulated symptom score is a mixture of autoregressive factors with unequal, timepoint-dependent variances, plus a participant baseline term. The M3 working covariance, a single random intercept and a single-parameter corCAR1 residual, cannot match a mixture of autoregressive processes exactly. Increasing the number of participants estimates the parameters of that working covariance ever more sharply, but it does not change the fact that the covariance being estimated is, in a sense, the wrong one, and so the model-based standard error converges to a biased target. The bias persists because it is a bias of the model, not of the estimate of the model's parameters.

4.4 A cluster-robust standard error restores calibration

[23]

The natural remedy is a cluster-robust SE that is consistent regardless of the working covariance; the CR2 bias-reduced form is chosen for the moderate-cluster regime.

Table 4: Cluster-robust remedy at the primary null cell (3000 replicates). The model-based standard error of the production model is replaced by a CR2 bias-reduced cluster-robust standard error clustered on participant, with Satterthwaite degrees of freedom. The CR2 standard-error ratio is approximately one and the type-I error approximately 0.05.

Spec	κ model-based	κ CR2	Type I model	Type I CR2	CR2 df
A1	1.062	0.978	0.036	0.048	23.7
A2	1.096	1.000	0.029	0.045	23.7
A3	1.053	0.976	0.039	0.049	23.7

- The corCAR1 model and its point estimate are retained; only the standard error is replaced.
- The CR2 estimator of Bell and McCaffrey targets the moderate-cluster setting and is implemented in `clubSandwich`.
- Satterthwaite degrees of freedom from Pustejovsky and Tipton provide the matching small-sample reference.

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The diagnosis points to a remedy that does not require abandoning the corCAR1 model. If the model-based standard error is biased because the working covariance is wrong, the natural replacement is a standard error that is consistent whatever the working covariance happens to be[5,6,32]. We replace the model-based standard error of the interaction with a cluster-robust sandwich standard error, clustered on participant, in the bias-reduced CR2 form of Bell and McCaffrey[8], with the Satterthwaite degrees of freedom of Pustejovsky and Tipton[10,17]. The CR2 adjustment and its associated small-sample test were designed for precisely the regime we face here, a moderate number of clusters, and are implemented for `lme` objects in the `clubSandwich` package[33,34].

[24]

The CR2 estimator uses adjustment matrices $A_i = (I - H_{ii})^{-1/2}$ to achieve exact unbiasedness under the working covariance, and Satterthwaite degrees of freedom are recovered from the eigenvalues of the defining quadratic form.

- CR0 is consistent but downward-biased at moderate cluster counts; the jackknife CR3 over-corrects.
- CR2 takes the symmetric square root of $(I - H_{ii})$, generalising the HC2 estimator of MacKinnon and White.

- For `lme` objects the construction is applied in the GLS-transformed metric following Pustejovsky and Tipton.

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It is worth setting down the construction. Let $\hat{e}_i$ collect the residuals of cluster i from the fitted model and let H_{ii} be the corresponding block of the hat matrix. A cluster-robust variance replaces the model-based variance of $\widehat{\theta}$ by a quadratic form in the empirical cluster cross-products,

$$\widehat{V}_{\mathrm{CR}} \;=\; (X'X)^{-1}\Big(\textstyle\sum_i X_i' A_i\,\hat{e}_i\hat{e}_i'\,A_i X_i\Big)(X'X)^{-1}, \tag{10}$$

and the family CR0, CR1, CR2, CR3 is indexed by the adjustment matrices A_i[9,10]. The unadjusted choice $A_i = I$ is consistent as the number of clusters grows but biased downward at moderate cluster counts, and its own sampling variability is substantial in that regime[35,36]; the jackknife choice $A_i = (I - H_{ii})^{-1}$ over-corrects and is, in the marginal setting, the estimator of Mancl and DeRouen[7]. The CR2 choice sets $A_i = (I - H_{ii})^{-1/2}$, the symmetric square root that renders $\widehat{V}_{\mathrm{CR}}$ exactly unbiased for the true variance when the working covariance is in fact correct; it is the cluster generalisation of the HC2 estimator of MacKinnon and White[37]. For a mixed model fitted by `lme` the construction is applied in the metric of the working covariance, that is, to the generalized-least-squares-transformed design and residuals, following Pustejovsky and Tipton[10], and is recommended with a matching small-sample reference by Imbens and Kolesar[38]. Because $\widehat{V}_{\mathrm{CR}}$ is itself a quadratic form in approximately normal residuals, its null distribution is approximated by a scaled χ^2, and the Satterthwaite degrees of freedom are recovered by matching its first two moments, $\nu = (\sum_g \lambda_g)^2 / \sum_g \lambda_g^2$ in the eigenvalues λ_g of the matrix defining that form[10,17].

[25]

The CR2 test restores the SE ratio to approximately one and the type-I error to approximately 0.05; Satterthwaite degrees of freedom of approximately 24 confirm the small-sample reference is doing real work.

- CR2 SE ratios are 0.97–1.00 versus 1.05–1.10 for the model-based test.
- The type-I error rises from approximately 0.03 to 0.045–0.049, within Monte Carlo error of 0.05.
- Satterthwaite df of approximately 24 (versus model-based 487) indicates that a naive normal reference would be mildly anti-conservative.

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Table 4 reports the verification. The model-based standard-error ratio reproduces the conservatism, between 1.05 and 1.10. The CR2 cluster-robust ratio is close to one, between 0.97 and 1.00, and the realised type-I error rises from approximately 0.03 under the model-based test to between 0.045 and 0.049 under the CR2 test, within Monte Carlo error of the nominal 0.05. The Satterthwaite degrees of freedom for the cluster-robust test are approximately 24, far below the model-based 487 and well below the participant count of 70, since they measure the heterogeneity of cluster leverage on the interaction contrast rather than a count of clusters, the effective number of clusters in the sense of Carter, Schnepel, and Steigerwald[38,39]; the small-sample reference is doing real work, and a naive normal reference for the sandwich statistic would itself be mildly anti-conservative[9,40].

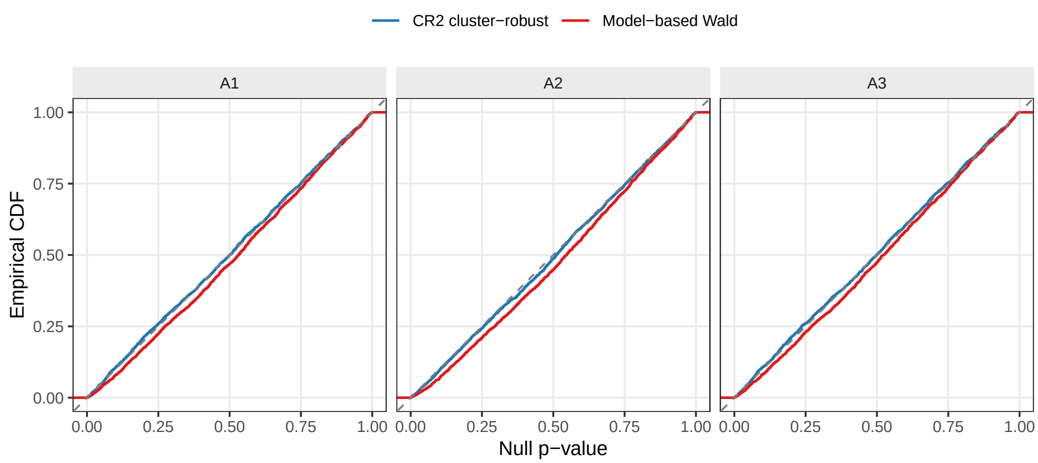

Figure 4: Null p-value distribution functions under the model-based and CR2 cluster-robust tests, by specification. The model-based curves bow below the diagonal; the CR2 curves lie on it.

[26]

The p-value CDF confirms that CR2 restores calibration without altering the point estimate or the analysis model.

- Model-based p-values bow below the diagonal consistently across all three specifications.
- Cluster-robust p-values lie on the Uniform(0,1) diagonal.
- Only the standard error and its reference distribution are replaced; the fitted model is otherwise unchanged.

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Figure 4 shows the same result as a distribution function. The model-based p-values bow below the diagonal, as in Figure 1; the cluster-robust p-values lie on it. The point estimate and the corCAR1 model are unchanged; only the standard error of the interaction, and the reference against which it is tested, are replaced.

4.5 The remedy at an alternative cell

[27]

The comparative ranking of specifications is checked at one alternative cell to determine whether recalibration disturbs their order.

- The alternative cell uses $c_{bm} = 0.45$, the parent study's reference non-null configuration.
- The question is whether recalibration changes not only absolute power but the ordering of specifications.
- The Architecture-B Hybrid design at $N = 70$ is retained for comparability with the null diagnosis.

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The diagnosis to this point has been conducted under the null, where calibration is defined. A reader of a comparative study will, however, press a further question. The purpose of the parent study is to rank analysis specifications by power, and if a recalibration of the standard error raises the power of every specification, does it also disturb their order? We checked this directly at one alternative cell, the parent study's reference configuration with a non-null biomarker moderation, $c_{bm} = 0.45$, retaining the Architecture-B Hybrid design at $N = 70$.

[28]

The recalibration acts as a near-uniform level shift that raises absolute power without disturbing the ranking of specifications.

- At the alternative cell the model-based ratio is 1.12–1.15; CR2

Table 5: The cluster-robust remedy at an alternative cell (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, 3000 replicates). Power is the rejection rate of the interaction test; the standard-error ratio κ is as defined for the null. The model-based test is conservative here too, and the CR2 test recalibrates it.

Spec	κ model-based	κ CR2	Power model	Power CR2
A1	1.124	0.986	0.757	0.795
A2	1.152	0.988	0.843	0.878
A3	1.119	0.989	0.755	0.788

brings it to 0.99 for all three specifications.

- Power rises by roughly three to four percentage points across specifications.
- The gaps between specifications are preserved: A2's advantage over A1 and A3 is essentially unchanged.

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Three things follow from Table 5. First, the standard-error over-estimation is not confined to the null: at the alternative cell the model-based ratio is 1.12 to 1.15, if anything slightly larger than the 1.06 to 1.10 seen under the null. Second, the CR2 cluster-robust standard error recalibrates the alternative cell as it did the null, bringing the ratio to 0.99 for all three specifications. Third, and most consequential for a comparative study, the recalibration is close to a uniform level shift. It raises the power of every specification by roughly three to four percentage points, and the gaps between specifications are left intact: the advantage of the leading specification A2 over A1 is $+0.086$ *under the model-based test and* $+0.082$ *under the cluster-robust test, and over A3 it is* $+0.088$ *and* $+0.089$*. The ranking, and the magnitude of the leading specification's advantage, survive the recalibration unchanged.*

[29]

A comparative study whose claim is a ranking is robust to recalibration because the correction shifts the level without reordering the contrast.

- The cluster-robust correction acts on the standard error, not on the point estimate.
- Because the mis-calibration is similar across specifications, the shift is approximately uniform.

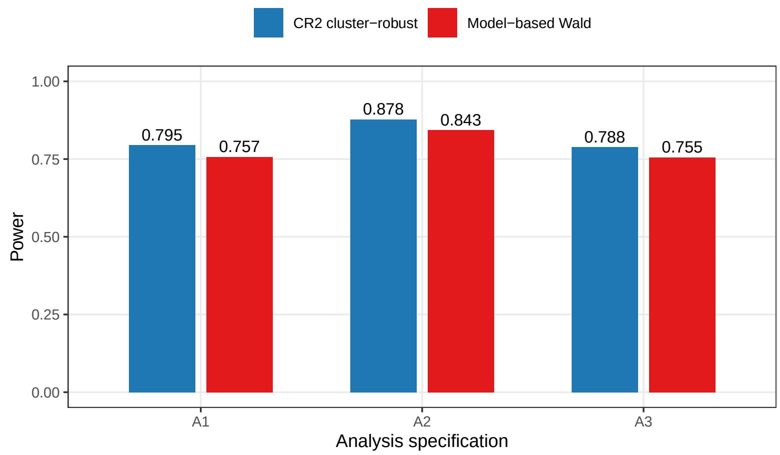

Figure 5: Power of the three analysis specifications at the alternative cell, under the model-based and CR2 cluster-robust tests. The recalibration raises every specification by a near-uniform increment and preserves their order.

- Absolute power figures reported under the model-based test are mildly pessimistic but the ranking is unaffected.

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The lesson generalises beyond the example. A cluster-robust standard error corrects the absolute error rate, the type-I error under the null and the power under the alternative alike, but because the correction acts on the standard error and not on the point estimate, and because the standard-error miscalibration here is relatively similar across the specifications being compared, it shifts the level without reordering the contrast. A comparative study whose claim is a ranking is therefore robust to the recalibration, even where its absolute power figures, as reported under the model-based test, are mildly pessimistic.

5 Discussion

5.1 The result is not about autoregressive correlation

[30]

The mechanism is general: any parametric working covariance capable of mismatching the truth is capable of miscalibrating the interaction test.

- The corCAR1 term is the component implicated here, but the model ladder shows four different ratios for four different working covariances.

- Mis-calibration from a wrong working covariance is the Eicker-Huber-White result; the worked example adds quantification and decomposition.
- The specific contribution is a realistic longitudinal case with two non-obvious features.

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It would be a mistake to read this paper as a caution against the continuous-time autoregressive residual in particular. The corCAR1 term is the component the model ladder happened to implicate in this example, but the mechanism is general. The model ladder is itself the evidence: it fitted four working covariances and obtained four different standard-error ratios, from 1.3 under ordinary least squares to 1.0 under a random intercept to near 1.1 under the production model. The mis-calibration is a property of the match between the working covariance and the truth, and any parametric residual or random-effects structure, compound symmetry, an unstructured form, a random slope, a spatial correlation, is capable of producing it; the effect of a serial-correlation structure on the variance of a least-squares contrast was examined classically by Watson[41], and the robustness of mixed-model fixed-effect inference to a misspecified covariance by Verbeke and Lesaffre[42]. That a mis-specified working covariance leaves the model-based fixed-effect standard error inconsistent is, moreover, not a new claim; it is the Eicker-Huber-White result carried into the mixed-model setting, and for a likelihood fit the misspecified-maximum-likelihood result of White[3–5,26,43]. What the worked example contributes is a quantified, decomposed instance in a realistic longitudinal interaction test, with two features that are not obvious in advance.

5.2 Two features that are not obvious in advance

[31]

For a within-cluster contrast the direction of the standard-error bias can be conservative rather than anti-conservative, reversing the usual sandwich-literature intuition.

- The sandwich literature is typically motivated by anti-conservative standard errors from ignored between-cluster correlation.
- Here the model-based test is conservative: the corCAR1 term, not the absence of modelling, introduced the bias.
- An analyst who infers conservatism from having modelled the correlation has the logic backward for a within-cluster contrast.

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The first is the direction. The sandwich literature is usually motivated by anti-conservative tests: ignored positive correlation under-states standard errors, and the robust estimator corrects an inflated rejection rate downward. Here the model-based test is conservative, and the cluster-robust estimator corrects the rejection rate upward. The ladder explains why the intuition can mislead. The effect under test is a within-cluster contrast, and for a within-cluster contrast the relationship between a mis-specified covariance and the standard-error bias need not have the sign that the between-cluster intuition supplies. An analyst who reasons that 'my standard errors are conservative because I have modelled the correlation' has the logic backward in this case: modelling the correlation, through the corCAR1 term, is what introduced the conservatism, and the less elaborate model was the calibrated one.

[32]

Kenward-Roger corrections address finite-sample artefacts and are the wrong instrument here because the bias is asymptotic and does not attenuate with sample size.

- An earlier informal reading attributed the conservatism to moderate sample size; the sample-size gradient refutes this.
- Kenward-Roger corrections operationalise the Kackar-Harville result for finite-sample variance underestimation.
- An asymptotic mismatch is not corrected by Kenward-Roger; the cluster-robust SE addresses it directly.
- Applied empirically through `mmrm`, Kenward-Roger is inert on the one-parameter autoregressive covariance yet corrects a real finite-sample anti-conservatism on an unstructured one.

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The second feature is that the bias is asymptotic. It would have been straightforward to attribute the conservatism to moderate sample size, and an earlier informal reading of the parent study did so. The sample-size gradient refutes that reading directly. This matters for how the problem is classified. The small-sample corrections of Kenward and Roger[23,27] adjust the

Table 6: Kenward-Roger versus Satterthwaite degrees of freedom at the primary null cell (2000 replicates), applied through the `mmrm` package to two marginal working covariances: a continuous-time autoregressive (spatial-exponential) covariance, the marginal counterpart of the corCAR1 residual layer, and an unstructured covariance. Values are averaged over the three specifications. κ is the mean model standard error over the empirical standard deviation of the estimate. On the one-parameter autoregressive covariance the Kenward-Roger correction is inert; on the thirty-six-parameter unstructured covariance it corrects a genuine finite-sample anti-conservatism. Computed with the mmrm package (version 0.3.17).

Covariance	Method	κ	Mean df	Type I
AR(1), 1 parameter	Satterthwaite	1.075	476	0.036
AR(1), 1 parameter	Kenward-Roger	1.076	476	0.036
Unstructured, 36 parameters	Satterthwaite	0.892	164	0.076
Unstructured, 36 parameters	Kenward-Roger	0.955	164	0.059

fixed-effect standard error and its degrees of freedom for the finite-sample uncertainty in the estimated variance parameters, operationalising the variance-underestimation result of Kackar and Harville[28]; they are the right instrument for a finite-sample artefact. They are not the right instrument here, because the bias is not a finite-sample artefact: it does not vanish as the variance parameters are estimated with unlimited precision. A Kenward-Roger correction applied to a mis-specified working covariance corrects the finite-sample term and leaves the asymptotic mismatch untouched. We confirmed this directly. Because the Kenward-Roger correction is not implemented for the production corCAR1 mixed model in standard software, we applied it through the `mmrm` *package[44] to the marginal counterpart of the implicated autoregressive layer, a continuous-time spatial-exponential working covariance, with an unstructured covariance fitted alongside for contrast (Table 6). On the autoregressive covariance the correction was numerically inert: it altered the interaction standard error by about one tenth of one percent, holding the standard-error ratio at* $\kappa \approx 1.08$ *and the type-I error near 0.036, because a single correlation parameter carries almost no finite-sample uncertainty for the correction to absorb. On the thirty-six-parameter unstructured covariance the same correction was far from inert, lifting the standard-error ratio from 0.89 to 0.96 and the type-I error from 0.076 toward 0.059. The magnitude of the Kenward-Roger adjustment thus tracks the number of covariance parameters, and for a parsimonious autoregressive working covariance it is too small to offset a six-to-ten percent bias. The cluster-robust standard error addresses the asymptotic mismatch directly, which is why it is the appropriate remedy.*

5.3 When the problem should be expected

[33]

The conditions for this problem are met by nearly every applied mixed-model interaction test, and routine accompani-

ment by a calibration check or cluster-robust SE is warranted.

- The conditions are: a fixed-effect interaction, a parametric working covariance not known to be exact, and a within-cluster contrast.
- These conditions characterise almost every applied mixed-model interaction analysis.
- A calibration check or cluster-robust SE removes the question at modest additional cost.

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The conditions for the problem are not exotic. They are a fixed-effect interaction tested in a mixed model; a parametric working covariance that is plausible but not known to be exactly correct, which is to say almost every parametric working covariance in applied use; and an interaction that is partly or wholly a within-cluster contrast. Wherever those conditions hold, the calibration of the model-based Wald test is an open question, and the staged protocol of Section 2 answers it at the cost of a simulation. We would suggest, as a matter of routine practice, that an interaction test reported from a mixed model with a non-trivial working covariance be accompanied either by a calibration check of this kind or by a cluster-robust standard error, which removes the question by being agnostic to the working covariance. This accords with a broader case for giving robust and marginal methods more routine consideration alongside the conditional mixed model[45].

5.4 The marginal-model alternative

[34]

GEE is a natural alternative that carries the sandwich variance from the outset; for a linear model with identity link it estimates the same interaction as the conditional mixed model.

- Liang-Zeger GEE decouples point estimation from variance estimation by design.
- For a linear identity-link model, marginal and conditional fixed-effect coefficients coincide.
- The working correlation governs efficiency, not inferential validity.

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Table 7: Marginal (GEE) check at the primary null cell, with an AR(1) working correlation (5000 replicates). The naive sandwich standard-error ratio κ and type-I error are shown beside the Mancl-DeRouen (MD) bias-corrected versions. The naive sandwich is mildly anti-conservative; the bias correction restores approximate calibration.

Spec	κ naive	κ MD	Type I naive	Type I MD
A1	0.955	0.992	0.068	0.055
A2	0.963	0.994	0.067	0.058
A3	0.955	0.991	0.069	0.055

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The remedy adopted above keeps the conditional mixed model and adds a cluster-robust standard error after the fact. A reader may reasonably ask whether the marginal route, generalized estimating equations, would not be more natural, since the method carries the sandwich variance from the outset rather than bolting it on[6,46]. It would. For a linear model with an identity link the marginal and conditional fixed-effect coefficients coincide[47], so a generalized estimating equations analysis estimates the same interaction and answers the same question; the working correlation, exchangeable or autoregressive, governs only the efficiency of the point estimate and not the validity of the inference.

[35]

The GEE route is viable but requires its own small-sample correction; both the conditional CR2 and marginal Mancl-DeRouen tests are defensible calibrated analyses and the choice is practical rather than principled.

- The naive GEE sandwich under-estimates the sampling SD by about four percent, yielding a type-I error of 0.067–0.069.
- Mancl-DeRouen bias correction brings the ratio to within one percent of one and the type-I error to 0.055–0.058.
- The conditional CR2 test is slightly conservative; the marginal Mancl-DeRouen test slightly anti-conservative; both are defensible.

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We checked the marginal route at the primary null cell, fitting the generalized estimating equations counterpart of the analysis model with an AR(1) working correlation; Table 7 reports the result. The naive sandwich standard error proved mildly anti-conservative, the small-sample behaviour the literature describes for a moderate number of clusters[36,48]: across the three specifications it under-estimated the sampling standard deviation by roughly four percent, and the realised type-I error was 0.067 to 0.069. A bias-corrected sandwich of the kind introduced by Mancl and DeRouen[7] removed most of the discrepancy, bringing the standard-error ratio to within one percent of one and the type-I error to 0.055 to 0.058. The marginal route is therefore viable, but it carries the same lesson as the conditional one: at the cluster counts typical of aggregated N-of-1 work the sandwich variance itself needs a small-sample correction[35]. The two corrected tests err in opposite and relatively modest directions, the conditional CR2 test slightly conservative and the marginal Mancl-DeRouen test slightly anti-conservative, and either is a defensible calibrated analysis. The choice between them is practical rather than principled: the conditional route preserves an existing mixed-model pipeline and changes only the standard error, while the marginal route is conceptually cleaner but a more thoroughgoing change of model.

5.5 Limitations

[36]

The scope is bounded to one data-generating family, one estimand, and one working covariance, and the power consequence is verified at only a single alternative cell.

- Generality beyond the corCAR1-plus-random-intercept model rests on the model ladder and established theory, not on an empirical sweep.
- The gradient varied the number of clusters but not the cluster size; a gradient in timepoints per participant is a further extension.
- The CR2 remedy spends precision (Satterthwaite df approximately 24) to buy validity, and that cost rises at smaller cluster counts.
- Whether the near-uniform power shift holds across the full factorial rather than the one cell examined remains an open question.

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Several limitations bound the scope of the evidence, and we shall endeavour to state them plainly. The diagnosis was carried out for one data-generating *family, one estimand, and one production working covariance; the asymptotic conservatism is demonstrated for the corCAR1-plus-random-intercept model, and the generality beyond that rests on the model ladder and on established theory rather than on a sweep of every covariance structure. A broader empirical base, fitting an unstructured or compound-symmetry analysis to the same data, would strengthen the general claim, and is a natural extension. In the same spirit, the sample-size gradient varied the number of participants while holding the within-participant design fixed; the number of timepoints per cluster, the axis along which the per-cluster working-covariance mismatch could itself grow or shrink, was not varied, and a gradient in cluster size is a further natural extension. The cluster-robust remedy is not without cost: the Satterthwaite degrees of freedom for the CR2 test are approximately 24 against a model-based 487, so the robust test spends precision to buy validity, and at a genuinely small number of clusters that price would rise. The principal diagnosis was conducted at the null, where calibration is defined; the consequence for power was checked at a single alternative cell (Section 4.5), where the recalibration proved a near-uniform level shift that preserved the ranking of specifications. Whether that uniformity holds across the whole factorial, rather than at the one cell examined, was not established, and that question warrants further investigation. Finally, the small positive point-estimate bias noted in Section 4.1, while negligible in magnitude and irrelevant to the conservatism, is a genuine finite-sample feature of the estimator that a separate examination might characterise.*

6 Conclusions

In conclusion, six points are to be emphasised. First, the model-based Wald test of a fixed-effect interaction in a linear mixed model can be materially mis-calibrated when the parametric working covariance does not match the data-generating covariance. In the worked example the mis-calibration is a conservative one: the model-based standard error of the interaction is over-estimated by 6 to 10 percent and the realised type-I error is approximately 0.03 against a nominal 0.05.

Second, the four-channel decomposition localises the discrepancy to the standard-error channel and excludes point-estimate bias, the degrees-of-freedom rule, and the positive-definiteness correction.

Third, the working-covariance model ladder localises the cause to the corCAR1 residual layer. A random intercept alone is well calibrated; adding the corCAR1 residual removes the calibration.

Fourth, the sample-size gradient shows the mis-calibration is asymptotic. It does not attenuate from $N = 35$ to $N = 280$, and so it is a property of the working covariance and not a small-sample artefact amenable to a Kenward-Roger correction.

Fifth, a CR2 bias-reduced cluster-robust standard error, clustered on participant and referred to a Satterthwaite distribution, restores the standard-error ratio to approximately one and the type-I error to approximately 0.05, without altering the point estimate or the analysis model. At an alternative cell the same recalibration is a near-uniform level shift in power that preserves the comparative ranking of specifications.

Sixth, and in our view most consequential for routine practice, the staged protocol, comprising the four-channel decomposition, the model ladder, and the sample-size gradient, is a reusable instrument for any mixed-model interaction test whose calibration is in doubt. We recommend that interaction tests reported from mixed models with a non-trivial working covariance be accompanied by a calibration check of this kind, or by a cluster-robust standard error that is agnostic to the working covariance.

7 Reproducibility

[37]

All diagnostic outputs are generated by seven named scripts and read inline; the manuscript is fully reproducible from the repository root.

- Seven scripts under `analysis/scripts/carryover-sensitivity/` cover the decomposition, model ladder, sample-size gradient, null and alternative CR2 verifications, the GEE check, and the Kenward-Roger check.
- Each script writes a replicate-level `.rds` file; this manuscript reads those files and regenerates every table and figure inline.
- The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

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The diagnosis is driven by seven scripts under `analysis/scripts/carryover-sensitivity/`: `diagnose-null-type1.R` for the four-channel decomposition, `diagnose-null-type1-stages46.R` for the model ladder and the sample-size gradient, `diagnose-null-type1-cr2.R` for the cluster-robust verification at the null, `diagnose-alt-cell-cr2.R` for the alternative-cell check, `diagnose-null-type1-gee.R` for the GEE marginal-model check, and `diagnose-null-type1-kr.R` for the Kenward-Roger check, with `check-null-architecture-equivalence.R` establishing the architecture-invariance of the null. Each writes a replicate-level `.rds` file into the `output/` subdirectory, and this manuscript reads those files and regenerates every table and figure inline. The cluster-robust standard errors use the `clubSandwich` package, and the Kenward-Roger and Satterthwaite checks use the `mmrm` package[44]. The examination plan that preceded the manuscript is retained alongside it as `type1-examination-plan.md`.

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